

Zhenhao Zhang

New York, NY | (585) 470-3118 | zz3530@columbia.edu | LinkedIn | GitHub
AI Agents · Multimodal Learning · Reinforcement Learning · Computer-Use Agents

EDUCATION

Columbia University

New York, NY

M.S. in Artificial Intelligence; Concentration: AI and Advanced Computing

Aug. 2026 – Dec. 2028 (Expected)

University of Rochester

Rochester, NY

B.S. in Computer Science; B.S. in Business: Information Systems

Aug. 2022 – Dec. 2025

- GPA: **3.85/4.00**; Cum Laude (Computer Science & Business Information Systems); Highest Distinction in Computer Science; Dean's List (2023–2025)

SELECTED PUBLICATIONS

Mitigating Conversational Inertia in Multi-Turn Agents

ICML 2026

SeekJudge: A Practical Reward Framework for Reinforcement Learning in Computer-Use Agents

arXiv, 2026

Political Symbols and Urban Amenities: The Spatial Logic of Flag Placement in Istanbul

SSRN, 2026

RESEARCH EXPERIENCE

Zhejiang University, College of Computer Science and Technology

Hangzhou, China / Hybrid

Researcher; Supervisor: Dr. Linchao Zhu

Feb. 2025 – Present

Computer-Use CAPTCHA Agents | *Lead Researcher*

Jan. 2026 – Present

- Led development of an interactive multimodal benchmark and training environment with **50K CAPTCHA instances** across **20 task categories** and 5 interaction patterns, using mouse/keyboard actions and executable browser-verified ground truth.
- Designed an answer-conditioned reasoning distillation pipeline for **46K multi-turn trajectories**, integrating step-level reasoning supervision, symbol-to-pixel grounding, persistent PLAN/STATE memory, and independent VLM quality filtering.
- Trained a unified **Qwen3.5-9B** multimodal agent across all 20 task categories on a 37K+ per-turn SFT corpus, experimenting with LoRA, full-parameter fine-tuning, frozen vision encoders, 32K contexts, and multi-task data scheduling.
- Engineered large-scale multimodal SFT/GRPO training infrastructure with DeepSpeed ZeRO-3, Liger FLCE, distributed checkpointing, executable environment rewards, and custom memory optimizations for long-context training with a 248K-token vocabulary.

SeekJudge | *Computer-Use Agent Reward Modeling* | *arXiv 2026*

Jan. 2026 – Aug. 2026

- Built and manually annotated a **300+ trajectory** cross-platform benchmark for long-horizon computer-use agents, with trajectory-level verdicts and dense step-level supervision across web, desktop, and mobile environments.
- Designed fine-grained reward criteria covering milestones, correct decisions, error correction, off-path behavior, and harmful actions; conducted human evaluation and model-vs.-human disagreement analysis at the step level.
- Performed systematic error analysis of rule-based and model-based reward signals, identifying rule-human disagreement across multiple CUA environments and motivating learned reward models for long-horizon agent evaluation.
- Contributed to distillation and SFT training of **SeekJudge-9B**, analyzing reasoning, formatting, and termination failure modes and iterating on training data and supervision strategies.

Agent Memory / Conversational Inertia | *ICML 2026*

Feb. 2025 – Aug. 2026

- Implemented task-specific trajectory summarization for long-horizon LLM agents across Maze, WebShop, Wordle, SciWorld, and TextCraft, compressing multi-turn observation-action histories while preserving task-critical state and progress.
- Designed environment-aware memory representations and summarization prompts, selectively retaining navigation state, completed subgoals, constraints, prior actions, and unresolved information for downstream agent decision-making.
- Benchmarked summarization against alternative context-management strategies across heterogeneous **AgentGym** environments, analyzing the trade-off between history retention and conversational inertia; training-free summary prompts improved Maze success from 85.5% to **90.0%**.
- Conducted failure and behavioral analysis of long-context agents, building attention diagnostics that link system-token ratio → system attention → task success, and studying action repetition, exploration, and error propagation.

INDUSTRY EXPERIENCE

Merit Interactive Co., Ltd. <i>AI Engineer Intern</i>	Hangzhou, China <i>Jan. 2026 – Mar. 2026</i>
- Designed an end-to-end multimodal retrieval system over 770K+ images, supporting text-to-image and image-to-image search through FAISS ANN retrieval, reranking, and score fusion, with <50 ms vector recall and ~100–500 ms end-to-end query latency. - Built a multilingual webpage classification pipeline across 8 topic domains; labeled ~6K pages, curated 7.5K+ samples, and benchmarked 13 approaches spanning embedding classifiers, contrastive learning, LoRA, clustering, and retrieval/reranking. - Improved classification from 62.5% to 87.75% Top-1 (97.6% Top-2) and deployed the final lightweight 264 KB model as a containerized REST API.	

Merit Interactive Co., Ltd. <i>Algorithm Intern</i>	Hangzhou, China <i>Jul. 2024 – Aug. 2024</i>
- Built mobile-app analytics datasets from raw user logs with SQL; analyzed usage, installation, uninstallation, user distribution, and temporal trends and automated Top-100 ranking reports. - Developed app-trend forecasting models using Logistic Regression and LightGBM with behavioral and temporal features to predict applications likely to trend in the following week.	

ADDITIONAL RESEARCH & TEACHING

University of Rochester, Goergen Institute for Data Science and AI <i>DSCC 279 Independent Research; Supervisor: Prof. Cantay Çalışkan</i>	Rochester, NY <i>Aug. 2025 – Jan. 2026</i>
- Built a geospatial vision pipeline over ~748K Google Street View panoramas, including image cleaning, coordinate alignment, spatial matching, multi-scale geographic aggregation, and district-level feature construction. - Applied nested geographic buffers, distance-weighted aggregation, and Earth Mover’s Distance to quantify structural differences in political-symbol visibility across urban districts.	
University of Rochester, Department of Computer Science <i>Undergraduate Teaching Assistant, CSC 280: Computer Models and Limitations</i>	Rochester, NY <i>Aug. 2025 – Dec. 2025</i>

SELECTED PROJECTS & AWARDS

GIDS Biomedical Data Science Hackathon MERFISH Cell-Type Classification 3rd/30	Aug. 2026
Led the modeling pipeline for 60-class cell-type prediction on mouse spinal-cord MERFISH data (200 genes, 5K labelled cells): seed-bagged LightGBM ensembles over expression, PCA, and spatial/expression-neighbour label features with biological constraint masks and a leakage-free CV protocol; reached 80.7% accuracy (majority-class baseline 14%).	
GIDS Biomedical Data Science Hackathon Perturb-seq Gene Expression Prediction	Aug. 2025
Developed an enhanced Norman-method pipeline for single/double perturbation prediction using robust preprocessing, multi-model regression, and adaptive coefficient estimation; achieved RMSD 0.40 on the test set.	
GIDS Biomedical Data Science Hackathon Single-Cell Multiome Modeling	Aug. 2024
Built an MLP-based peak-to-gene prediction pipeline for single-cell multiome data, achieving 75% accuracy and outperforming baseline linear and random-forest models.	
Finnov8 Hackathon Second Best Overall Award	Mar. 2024
Achieved an individual score of 0.67 in a 4-member team and improved prediction from 0.60 to 0.67 using feature engineering, Multi-Output MLP, and LSTM models.	

LEADERSHIP

Minister of Publicity , University of Rochester CSSA	Jun. 2024 – Jul. 2025
Member, External Relations Department , University of Rochester CSSA	Jun. 2023 – May 2024

TECHNICAL SKILLS

<i>Languages:</i> Python, C/C++, Java, SQL, JavaScript/HTML/CSS	
<i>ML/AI:</i> PyTorch, Hugging Face Transformers, Scikit-learn, FAISS, LoRA, SFT, GRPO, multimodal/VLM training	
<i>Systems/Tools:</i> DeepSpeed ZeRO-3, Docker, Linux, Git, REST APIs, distributed training and checkpointing	